NotifyHub

Architecture & Workflow

A centralized backend notification platform that delivers notifications reliably and asynchronously, decoupled from producer business logic — built on the transactional outbox pattern, RabbitMQ, and Server-Sent Events.

Java 21 · Spring Boot 3.4 | PostgreSQL | RabbitMQ | React 19 · Vite | Outbox pattern | SSE | API-key auth | Gmail SMTP

01 · OVERVIEW

What the system is

NotifyHub turns a single "please notify this user" request into a reliable, observable delivery — without blocking or coupling the caller's workflow.

Producers submit a notification intent over REST. NotifyHub persists it as the source of truth, hands delivery to an asynchronous pipeline, and reports the final state (`SENT` or `FAILED`) both in the database and live over Server-Sent Events. Delivery failures are retried and, once exhausted, dead-lettered — never lost, never silently dropped.

Design principles

- **Database is the source of truth** — state is persisted explicitly.
- **Asynchronous, non-blocking** — the caller returns immediately.
- **Failure-aware** — bounded retries then explicit dead-lettering.
- **Best-effort real-time** — SSE augments, never replaces, the DB.

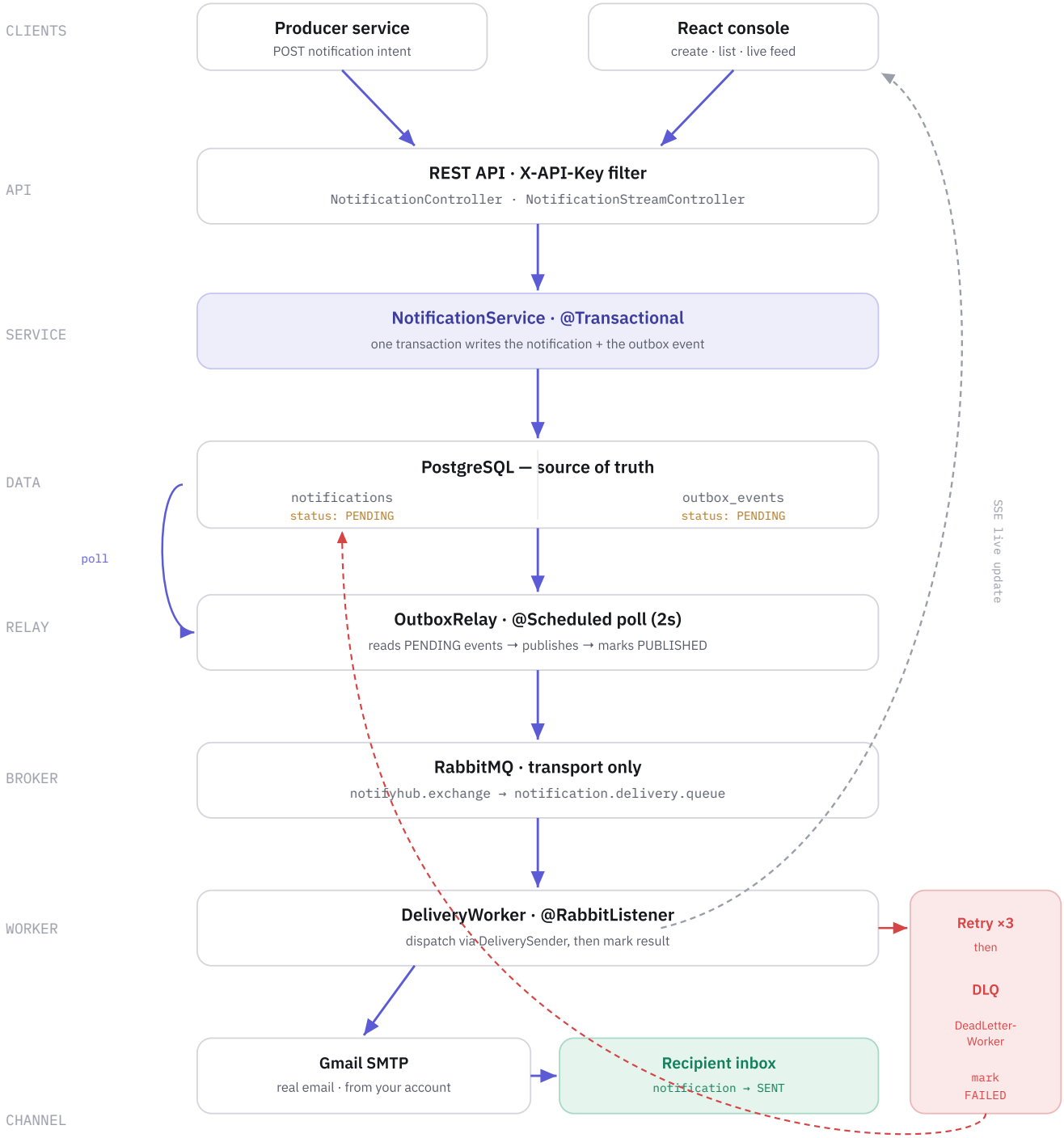
Guarantees

- **No lost events** — outbox write is in the same transaction as the notification.
- **At-least-once delivery** — the relay retries unpublished events.
- **Terminal states are recorded** — DLQ handler marks exhausted sends FAILED.
- **Transport ≠ state** — RabbitMQ carries work; Postgres holds truth.

02 · SYSTEM ARCHITECTURE

How a request travels across layers

Solid arrows = the delivery path. Dashed = live updates & the failure branch. The database sits at the center as the source of truth; RabbitMQ is transport only.



03 · REQUEST LIFECYCLE

End to end, step by step

What happens from the moment a producer calls `POST /api/v1/notifications`.

1. **Authenticate.** The `ApiKeyAuthFilter` checks `X-API-Key` (constant-time). Missing/invalid → `401`. Valid → the request proceeds.

2. **Validate.** `NotificationController` binds the JSON to a validated DTO (email format, required fields). Violations → `400` with an RFC-7807 body.

3. **Persist atomically.** `NotificationService.create()` runs in one transaction: it saves the `Notification` as `PENDING` *and* an `OutboxEvent` carrying the delivery payload. Both commit or neither does.

4. **Respond immediately.** The API returns `201 Created` with the notification — the caller is never blocked on delivery.

5. **Relay.** `OutboxRelay` (scheduled, every 2s) finds PENDING outbox rows, publishes each to RabbitMQ, and marks them `PUBLISHED`. A failed publish stays PENDING and is retried next tick — at-least-once.

6. **Consume & deliver.** `NotificationDeliveryWorker` (@RabbitListener) receives the task and calls the configured `DeliverySender` — a log mock or real Gmail SMTP.

7. **Record result.** On success the notification becomes `SENT`; the outcome is pushed to any connected SSE clients.

8. **Failure path.** If the send throws, RabbitMQ retries up to 3× with backoff; still failing, the message is dead-lettered. `DeadLetterWorker` marks it `FAILED` and notifies over SSE.

04 · RELIABILITY MODEL

Why nothing gets lost

Transactional outbox

Publishing to a broker and committing to a database can't share one transaction. Instead, the delivery event is written to `outbox_events` in the *same* DB transaction as the notification. A separate relay publishes it afterward. If the app crashes before publishing, the event is still there — it goes out on the next poll.

Retry & dead-letter

The listener retries a failing delivery 3× with exponential backoff. When attempts are exhausted the message is rejected to a dead-letter exchange and lands on `notification.delivery.dlq`, where `DeadLetterWorker` records the terminal `FAILED` state. Failures are visible, not silent.

CONCERN	MECHANISM	OUTCOME
Lost publish on crash	Outbox written in the notification's transaction	Event survives; relay resends
Broker/publish hiccup	Event stays PENDING, re-polled	At-least-once delivery
Provider outage	3× retry with backoff	Transient errors self-heal
Permanent failure	Dead-letter queue + handler	Recorded as FAILED
Slow SMTP host	10s connect/read/write timeouts	Fails fast into retry path

05 · COMPONENTS

Backend structure

PACKAGE	TYPE	RESPONSIBILITY
`notification.web`	Controllers / DTOs	REST intake, SSE stream, validation, RFC-7807 errors
`notification.service`	Service	Transactional create (notification + outbox), reads
`notification.domain`	Entities	`Notification`, `OutboxEvent` + status enums
`notification.repository`	Spring Data JPA	Persistence for notifications & outbox
`notification.outbox`	Scheduled relay	`OutboxRelay` — poll & publish to RabbitMQ
`notification.messaging`	Broker	Rabbit config, delivery worker, dead-letter worker
`notification.delivery`	Channel seam	`DeliverySender`: logging + Gmail SMTP impls
`notification.realtime`	SSE	`SseService`, per-recipient emitters
`config`	Security	API-key filter + stateless security chain

Frontend structure

PATH	CONTENTS
`src/lib`	Types, API client (API-key + RFC-7807 aware), formatters
`src/hooks`	`useNotifications` (TanStack Query), `useNotificationStream` (SSE)
`src/components`	DeliveryLane (signature view), NotificationsTable, ComposeForm, LiveFeed, NavRail, StatusPill

Stack: React 19 · Vite · TypeScript · Tailwind v4 · TanStack Query. Fonts: Space Grotesk, IBM Plex Sans/Mono.

06 · DATA MODEL & API

Persistence & endpoints

notifications

id uuid · primary key
to_email varchar(255)
subject varchar(255)
body text
status PENDING · SENT · FAILED
created_at · **updated_at** timestamp

outbox_events

id uuid · primary key
event_type varchar(100)
payload jsonb
status PENDING · PUBLISHED · FAILED
created_at · **published_at** timestamp
index on (status, created_at)

HTTP API

METHOD	PATH	PURPOSE	AUTH
POST	/api/v1/notifications	Create a notification intent	X-API-Key
GET	/api/v1/notifications	List notifications	X-API-Key
GET	/api/v1/notifications/{id}	Fetch one	X-API-Key
GET	/api/v1/notifications/stream	SSE delivery stream (per recipient)	apiKey query
GET	/actuator/health	Liveness probe	public

07 · SECURITY & DELIVERY

Access control and the send channel

API-key authentication

Every endpoint except the health probe requires a valid `X-API-Key` (the SSE stream accepts it as an `apiKey` query param, since EventSource can't set headers). Keys come from env, support comma-separated rotation, and are compared in constant time. Security is stateless — no sessions, CSRF disabled.

Scope: platform-level access. Per-recipient authorization (a user seeing only their own stream) would need user identity (JWT) — a documented next step.

Pluggable delivery

Delivery is a single seam, the `DeliverySender` interface, selected by `notifyhub.delivery.mode`:

`log` (default) — logs the dispatch; a subject containing "fail" exercises the DLQ path.
`smtp` — real email via Gmail (`smtp.gmail.com:587` , STARTTLS).

Gmail needs only `GMAIL_USER` + `GMAIL_APP_PASSWORD` (a 2FA App Password); sent mail's From is that account.

Runtime topology

Postgres and RabbitMQ run as Docker containers; the app runs locally (IDE) or as a third container. All configuration is env-driven with local defaults, so the app boots without a hand-written `.env` .

08 · WHAT WAS BUILT

Delivery log

Phases 2–4 of the platform plus the frontend, security, and real email — each increment committed and verified end-to-end.

COMMIT	CHANGE
`115081c`	Notification pipeline — intake, persistence, outbox, RabbitMQ worker, DLQ, SSE (Phases 2–4)
`8ca0c80`	Restructure into `backend/` + `frontend/` monorepo
`07765c3`	React delivery console (Vite + Tailwind + TanStack Query)
`6ade523`	Make datasource/rabbit config runnable from an IDE without env vars
`7511226`	Real email delivery via SMTP (pluggable `DeliverySender`)
`799952f`	Add `.env.example` template
`0206668`	Require API key on all endpoints
`b0d9c7e` → `b038d86`	SMTP hardening, then pin delivery to Gmail

Verified behavior

- Normal notification: `PENDING` → relay → worker → `SENT`.
- Fail path: subject with "fail" → retries → DLQ → `FAILED`.
- Real email delivered through SMTP to a live inbox (verified against a mail catcher).
- Auth matrix: no key → 401, wrong key → 401, valid key → 200/201, health public.